

EXP 2-2 Experimental Plan M14

Authors: Yu Zhuge, Bo Li, Shi Shu, Yan Lin, Huaiyu Zhou, Jiahui Liu

1. Aim of the Experiment

The aim of this experiment is to determine the shear modulus (G) of a metal and a polymer wire by analyzing the simple harmonic motion of a torsion pendulum created by the twisting of these wires.

The experimental plan is used to direct the experiment and predict the results with assumed values.

Also, during the experiment, we need to learn and practice according following aspects:

- 1). Plan and execute a set of experimental measurements to measure the shear modulus of the metal and polymer wire.
- 2). Use uncertainty strategy of calculated shear modulus in a scientific way.
- 3). Critically analyze and discuss the results and their implications on material properties.

2. Principle of the Experiment

The principle of the experiment is based on the physical behavior of materials under torsional stress, which leads to the oscillation of a torsion pendulum. The shear modulus (G) of a material, an intrinsic property that quantifies its response to shear stress, can be calculated through the equations,

$$G = \frac{4\pi LmR^2}{r^4 T^2} \text{ and } G = \frac{E}{2(1+\nu)},$$

where L is the length of the wire, r is the wire radius, m is the mass of a disc with radius R, T is the period of a complete oscillation cycle, E is the young's modulus, and ν is the Poisson's ratio.

3. Measurement Procedure ^[1]

- 1). Calibrate the instrument
- 2). Measure the diameter of the disk
- 3). Measure the weight of a disk with a balance.
- 4). Measure wire diameter
- 5). Detect the length of the polymer wire, then assemble the device to keep it in a vertical state to twist and detect the twisting angle, and detect the period and amplitude of the oscillation.
- 6). Detect the length of the metal wire, then assemble the device and keep it in vertical state to twist and detect the torsion angle, detect the period and amplitude of oscillation.
- 7). Induce a large angular displacement of the disc because the angle decreases with each loop and release it to start oscillation, measuring the period for 10 loops by timer, then calculate one period for one oscillation.

4. Estimating L and T

According to the assumed values and the formula of shear modulus, the relationship between length L and the period of oscillation T can be derived:

$$T^2 = \frac{4\pi m R^2}{r^4 G} L$$

Figure 1 shows the estimated T and L values of each material, and the graphs which show what the experiment

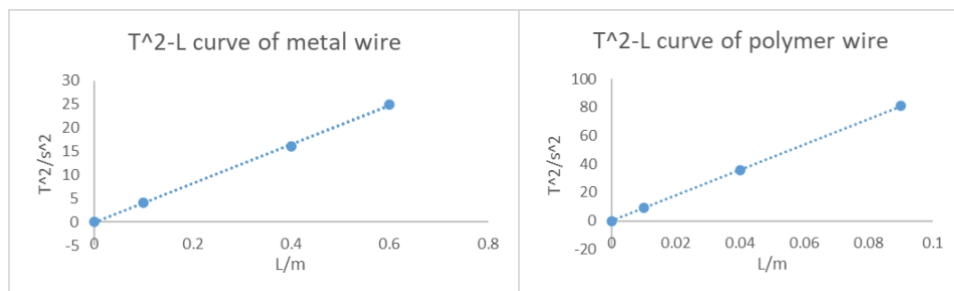


Figure 1. Estimated T and L values and the graphs to show what the results might look like.

might look like. For metal wire, $L=0.1\text{m}$, 0.4m , 0.6m ; $T=2\text{s}$, 4s , 5s . For polymer wire, $L=0.01\text{m}$, 0.04m , 0.09m ; $T=3\text{s}$, 6s , 9s .

5. Check the uncertainty.

The parameters caused the largest uncertainty will be the time period of oscillation (T) and the wire radius (r). The uncertainty caused by T is due to capturing the start and end of a cycle, and the extreme size of the wire would also introduce a lot of uncertainty.

6. The Values to be Measured.

Table 1 shows all the parameters that need to be measured. Among the parameters, amplitude and period of oscillation may contribute the largest uncertainty.

	Approximate Value		Experiment Equipment	Reading Resolution	Approximate Precision	Times
LENGTH (estimated)	Metal wire	100.0mm	Steel Measuring Tape	1 mm	1%	3 Times each length
		400.0mm				
		600.0mm				
	Polymer wire	10.0mm				
		40.0mm				
		90.0mm				
DIAMETER(R)	100 mm		Steel Measuring Tape	1 mm	1%	3 Different Directions 3 Times
DIAMETER(r)	Metal 0.6 mm Polymer 1mm		Micrometer	0.01 mm	Metal 1.7% Polymer 1%	3 Different Parts 3 Times
WEIGHT	1000 g		Balance	1 g	0.1%	3 Times
AMPLITUDE	200°		Protractor	1°	0.5%	3 Times
PERIOD of OSCILLATION (estimated)	Metal wire	2.00s	Stopwatch	0.01 s	1%	3 Times 4 cycles for each period
		4.00s				
		5.00s				
	Polymer wire	3.00s				
		6.00s				
		9.00s				

Table 1. All the estimated values and parameters that need to be measured.

7. References

1. QXU5017 - experiments in materials 2 - B23 - 2021 cohort [Internet]. [cited 2024 Mar 22]. Available from: <https://qmplus.qmul.ac.uk/course/view.php?id=24334>